

Spectral Equivariance and Geometric Transport in Reproducing Kernel Hilbert Spaces: A Unified Framework for Orthogonal Polynomial and Kernel Estimation

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Abstract

We develop a unified geometric framework for nonparametric estimation based on the notion of Twin Kernel Spaces, defined as orbits of a reproducing kernel under a group action. This structure induces a family of transported RKHS geometries in which classical orthogonal polynomial estimators, kernel estimators, and spectral smoothing methods arise as projections onto transported eigenfunction systems. Our main contribution is a Spectral Equivariance Theorem showing that the eigenfunctions of any transported kernel are obtained by unitary transport of the base eigenfunctions. As a consequence, orthogonal polynomial estimators are equivariant under geometric deformation, kernel estimators correspond to soft spectral filtering in a twin space, and minimax rates and bias–variance tradeoffs are invariant under transport. We provide examples based on Hermite and Legendre polynomials, affine and Gaussian groups, and illustrate the effectiveness of twin transport for adaptive and multimodal estimation. The framework reveals a deep connection between group actions, RKHS geometry, and spectral nonparametrics, offering a unified perspective that encompasses kernel smoothing, orthogonal series, splines, and multiscale methods.

Keywords: Reproducing kernel Hilbert spaces, orthogonal polynomials, spectral methods, group actions, nonparametric estimation, equivariance, minimax rates.

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1 Introduction

1.1 Motivation

Nonparametric estimation constitutes a cornerstone of modern statistical inference, with applications spanning density estimation, regression, and functional data analysis [Tsybakov, 2009, Wasserman, 2006]. Two classical paradigms have dominated this field: orthogonal series estimators based on polynomial expansions [Efremovich, 1999, Johnstone, 2017] and kernel smoothing methods [Wand and Jones, 1995, Silverman, 1986]. While these approaches appear methodologically distinct—the former relying on spectral truncation in a fixed basis, the latter on local averaging with bandwidth selection—both share deep connections to reproducing kernel Hilbert spaces (RKHS) [Berlinet and Thomas-Agnan, 2004, Cucker and Smale, 2002].

The present work is motivated by the observation that many seemingly different nonparametric methods can be understood as projections in different geometric “reference frames” of a common underlying RKHS structure. Specifically, we show that classical orthogonal polynomial systems—Hermite, Legendre, Laguerre, Jacobi, and their variants—arise naturally as eigenfunctions of transported kernels under group actions. This perspective unifies diverse estimation procedures and reveals fundamental geometric invariances that have not been systematically exploited in the statistical literature.

The theory of RKHS and Mercer kernels provides a natural framework for studying such connections [Aronszajn, 1950, Steinwart and Christmann, 2008]. Classical results on spectral decomposition of integral operators [Riesz and Sz.-Nagy, 1955, Dunford and Schwartz, 1988] establish that Mercer kernels admit eigenfunction expansions, which in the case of weighted L^2 spaces often coincide with classical orthogonal polynomials [Szegő, 1975, Ismail, 2005]. Our contribution is to show that this spectral structure is preserved—in a precise sense—under geometric transformations induced by group actions.

1.2 Contributions of the Paper

The main contributions of this paper are as follows:

- (i) **Twin Kernel Spaces framework.** We introduce the notion of Twin Kernel Spaces as orbits of a base reproducing kernel under a group action on the input space. This provides a minimal yet powerful geometric framework for studying families of related RKHS structures (Section 3).
- (ii) **Spectral Equivariance Theorem.** We prove that the eigenfunctions of any transported kernel are obtained by applying the unitary transport operator to the base eigenfunctions, with eigenvalues preserved. This result (Theorem 4.6) establishes that spectral structure is equivariant under geometric deformation.
- (iii) **Unified interpretation of estimators.** We demonstrate that orthogonal polynomial estimators correspond to hard spectral truncation while kernel smoothing corresponds to soft spectral attenuation, both operating in potentially different twin geometries. Bandwidth selection is reinterpreted as choice of twin space (Section 6).
- (iv) **Invariance of statistical properties.** We establish that convergence rates, bias–variance tradeoffs, and minimax risks are invariant under twin transport, providing a geometric explanation for the robustness of spectral methods (Sections 8–10).
- (v) **Explicit examples.** We provide detailed worked examples for Hermite polynomials with Gaussian geometry and Legendre polynomials with affine geometry, including a multimodal estimation example demonstrating the practical relevance of twin transport (Section 7).

1.3 Related Work

The connection between orthogonal polynomials and kernel methods has been explored from various angles. Smola et al. [2000] and Schölkopf and Smola [2002] discuss spectral properties of kernel operators in machine learning contexts. The role of orthogonal polynomials in density estimation dates to Cencov [1962] and was developed systematically by Efromovich [1999]. Wahba [1990] established foundational connections between spline smoothing and RKHS theory.

Group-theoretic perspectives on kernel methods appear in Kondor and Trivedi [2008] and Fukumizu et al. [2009], primarily in the context of invariant kernels for machine learning. The equivariance properties we establish differ in focusing on transport of spectral structure rather than invariance of the kernel itself.

The notion of “twin” structures in functional analysis has appeared in different contexts, notably twin spaces in operator theory [Halmos, 1982]. Our usage is distinct but shares the philosophy of paired geometric structures connected by canonical maps.

Recent work on adaptive estimation [Lepski et al., 1997, Goldenshluger and Lepski, 2011] and spatially inhomogeneous smoothing [Fan and Gijbels, 1996] addresses related concerns about geometry-dependent inference, though without the explicit group-theoretic framework we develop.

2 Orthogonal Polynomial Estimation Revisited

2.1 Orthogonal polynomial bases

Let $(E, \mathcal{E}, \mu)$ be a probability space with $E \subseteq \mathbb{R}$ an interval (bounded or unbounded) and μ a probability measure with finite moments of all orders. A sequence of polynomials $\{P_k\}_{k \geq 0}$ is *orthonormal* with respect to μ if $\deg(P_k) = k$ and

$$\int_E P_k(x) P_\ell(x) \mu(dx) = \delta_{k\ell}, \quad k, \ell \geq 0.$$

By the Gram–Schmidt process applied to the monomial basis $\{1, x, x^2, \dots\}$, such a system always exists and is unique up to sign [Szegő, 1975].

Classical examples include:

- **Hermite polynomials** $\{H_k\}_{k \geq 0}$: orthonormal with respect to the standard Gaussian measure $d\gamma(x) = (2\pi)^{-1/2} e^{-x^2/2} dx$ on $\mathbb{R}$.
- **Legendre polynomials** $\{P_k\}_{k \geq 0}$: orthonormal with respect to the uniform measure on $[-1, 1]$.
- **Laguerre polynomials** $\{L_k^{(\alpha)}\}_{k \geq 0}$: orthonormal with respect to the Gamma measure $x^\alpha e^{-x} dx$ on $[0, \infty)$.
- **Jacobi polynomials** $\{P_k^{(\alpha, \beta)}\}_{k \geq 0}$: orthonormal with respect to $(1-x)^\alpha (1+x)^\beta dx$ on $[-1, 1]$.

These systems satisfy three-term recurrence relations and have well-understood asymptotic properties [Ismail, 2005, Levin and Lubinsky, 2001].

2.2 Classical orthogonal series estimators

Given i.i.d. observations $X_1, \dots, X_n$ from a distribution with density f (with respect to μ), the classical orthogonal series density estimator is

$$\hat{f}_K(x) = \sum_{k=0}^K \hat{\theta}_k P_k(x), \quad \hat{\theta}_k = \frac{1}{n} \sum_{i=1}^n P_k(X_i), \quad (1)$$

where $K = K(n)$ is a truncation parameter [Efromovich, 1999, Wasserman, 2006].

Under standard regularity conditions, if $f = \sum_{k \geq 0} \theta_k P_k$ with $\sum_{k \geq 0} \theta_k^2 k^{2s} < \infty$ (Sobolev-type smoothness of order s), choosing $K \asymp n^{1/(2s+1)}$ yields the minimax optimal rate

$$\mathbb{E} \|\hat{f}_K - f\|_{L^2(\mu)}^2 = O(n^{-2s/(2s+1)}).$$

This rate is known to be unimprovable over Sobolev balls [Tsybakov, 2009].

2.3 RKHS interpretation

The orthogonal series estimator admits a natural RKHS interpretation. Define the Mercer kernel

$$K_e(x, y) = \sum_{k=0}^{\infty} \lambda_k P_k(x) P_k(y), \quad (2)$$

where $\{\lambda_k\}_{k \geq 0}$ is a summable sequence of positive eigenvalues (e.g., $\lambda_k = \rho^k$ for some $\rho \in (0, 1)$). The associated RKHS $\mathcal{H}_e$ consists of functions $f = \sum_k \theta_k P_k$ with $\sum_k \theta_k^2 / \lambda_k < \infty$, equipped with inner product

$$\langle f, g \rangle_{\mathcal{H}_e} = \sum_{k=0}^{\infty} \frac{\theta_k \phi_k}{\lambda_k}, \quad f = \sum_k \theta_k P_k, \quad g = \sum_k \phi_k P_k.$$

The integral operator $T_e : L^2(\mu) \rightarrow L^2(\mu)$ defined by

$$(T_e f)(x) = \int_E K_e(x, y) f(y) \mu(dy)$$

is compact, self-adjoint, and positive, with eigenpairs (λ_k, P_k) .

The truncated estimator (1) can be written as

$$\hat{f}_K = S_{e,K} \hat{f}_{\text{emp}}, \quad S_{e,K} f = \sum_{k=0}^K \langle f, P_k \rangle P_k,$$

where $\hat{f}_{\text{emp}}$ represents the empirical measure and $S_{e,K}$ is the spectral projection onto the first $K + 1$ eigenfunctions. This establishes that orthogonal series estimation is fundamentally a spectral truncation operation in an RKHS geometry [Berlinet and Thomas-Agnan, 2004, Cucker and Smale, 2002].

3 Twin Kernel Spaces: A Minimal Framework

3.1 Group actions on the input space

Definition 3.1 (Twin Space). *A twin space associated with (E, G) is the orbit $G \cdot E$, equipped with the induced structure.*

More precisely, let G be a group acting measurably on E via $\tau : G \times E \rightarrow E$, $(g, x) \mapsto g \cdot x$. We assume throughout that for each $g \in G$, the map $\tau_g : x \mapsto g \cdot x$ is a measurable bijection of E .

Typical examples include:

- The *affine group* $G = \{(a, b) : a > 0, b \in \mathbb{R}\}$ acting on $E = \mathbb{R}$ by $g_{a,b} \cdot x = ax + b$.
- The *multiplicative group* $G = (0, \infty)$ acting on $E = \mathbb{R}$ by $g_\alpha \cdot x = \alpha x$ (dilations).
- The *translation group* $G = \mathbb{R}$ acting on $E = \mathbb{R}$ by $g_b \cdot x = x + b$.

3.2 Transport of functions

Definition 3.2 (Twin Transform). *Given $f : E \rightarrow \mathbb{R}$, define the transported function $(g_1, g_2) \cdot f$ by*

$$(g_1, g_2) \cdot f(x) = g_2 \cdot f(g_1 \cdot x).$$

In our context, we focus on the case where the action on the function space is determined by the geometric action through change of variables with appropriate Jacobian correction.

3.3 Twin Kernel Spaces

Definition 3.3 (Twin Kernel Space). *A Twin Kernel Space is the family of kernels*

$$K_g(x, y) = g_2 \cdot K_e(\bar{g}_1 \cdot x, \bar{g}_1 \cdot y)$$

generated from a base kernel K_e , where $\bar{g}_1$ denotes the inverse action.

The precise form of g_2 is determined by requiring the transported kernel to define a valid reproducing kernel in the same $L^2(\mu)$ space, which necessitates Jacobian correction factors as detailed in Section 4.

4 Spectral Equivariance Theorem

In this section we formalize the notion of spectral equivariance in Twin Kernel Spaces and show how orthogonal polynomial systems arise as eigenfunctions of transported kernels. We work in a general L^2 -framework and specialize to orthogonal polynomials when needed.

4.1 Setting and assumptions

Let $(E, \mathcal{E}, \mu)$ be a σ -finite measure space and denote by

$$L^2(\mu) = \left\{ f : E \rightarrow \mathbb{R} \mid \int_E f(x)^2 \mu(dx) < \infty \right\}$$

the usual Hilbert space with inner product

$$\langle f, g \rangle_{L^2(\mu)} = \int_E f(x)g(x) \mu(dx).$$

Let $K_e : E \times E \rightarrow \mathbb{R}$ be a symmetric, measurable, positive definite kernel such that the associated integral operator

$$(T_e f)(x) = \int_E K_e(x, y) f(y) \mu(dy), \quad f \in L^2(\mu), \quad (3)$$

is Hilbert–Schmidt (in particular, compact and self-adjoint). Then there exists an orthonormal basis $\{P_k\}_{k \geq 0}$ of $L^2(\mu)$ and non-negative eigenvalues $\{\lambda_k\}_{k \geq 0}$ such that

$$T_e P_k = \lambda_k P_k, \quad k \geq 0, \quad (4)$$

and the kernel admits the convergent spectral expansion

$$K_e(x, y) = \sum_{k \geq 0} \lambda_k P_k(x) P_k(y) \quad \text{in } L^2(\mu \otimes \mu). \quad (5)$$

In the classical orthogonal polynomial setting, the functions P_k form an orthonormal polynomial system with respect to μ .

We now introduce a group action on E and the corresponding transport operators.

Definition 4.1 (Group action and Jacobian). *Let G be a group and let*

$$\tau : G \times E \rightarrow E, \quad (g, x) \mapsto g \cdot x,$$

be a measurable left action of G on E . For each $g \in G$, we write $\tau_g(x) = g \cdot x$. We assume that for every $g \in G$, the pushforward measure $\mu \circ \tau_g^{-1}$ is absolutely continuous with respect to μ , and we denote by

$$J_g(x) = \frac{d(\mu \circ \tau_g^{-1})}{d\mu}(x)$$

its Radon–Nikodym derivative (Jacobian).

The following operator implements the action of G on $L^2(\mu)$ in a unitary way.

Definition 4.2 (Twin transport operator on L^2). *For each $g \in G$, define $U_g : L^2(\mu) \rightarrow L^2(\mu)$ by*

$$(U_g f)(x) = J_g(x)^{1/2} f(g^{-1} \cdot x). \quad (6)$$

Lemma 4.3 (Unitarity of U_g). *For each $g \in G$, the operator U_g is unitary on $L^2(\mu)$, that is,*

$$\langle U_g f, U_g h \rangle_{L^2(\mu)} = \langle f, h \rangle_{L^2(\mu)} \quad \text{for all } f, h \in L^2(\mu),$$

and $U_{g_1 g_2} = U_{g_1} U_{g_2}$, $U_{e_G} = \text{Id}$.

Proof. Let $f, h \in L^2(\mu)$ and fix $g \in G$. Then

$$\langle U_g f, U_g h \rangle = \int_E J_g(x)^{1/2} f(g^{-1} \cdot x) J_g(x)^{1/2} h(g^{-1} \cdot x) \mu(dx) = \int_E J_g(x) f(g^{-1} \cdot x) h(g^{-1} \cdot x) \mu(dx).$$

Perform the change of variables $y = g^{-1} \cdot x$, so that $x = g \cdot y$. By the pushforward identity $\tau_{g*} \mu = J_g \mu$ (Definition 4.1), for any integrable Φ one has $\int_E \Phi(g \cdot y) \mu(dy) = \int_E \Phi(x) J_g(x) \mu(dx)$; equivalently

$$\mu(dx) = J_g(g \cdot y)^{-1} \mu(dy),$$

so that the integrand factor cancels: $J_g(x) \mu(dx) = J_g(g \cdot y) J_g(g \cdot y)^{-1} \mu(dy) = \mu(dy)$. Hence

$$\langle U_g f, U_g h \rangle = \int_E f(y) h(y) \mu(dy) = \langle f, h \rangle.$$

The group property $U_{g_1 g_2} = U_{g_1} U_{g_2}$ follows from a direct computation using the chain rule for Radon–Nikodym derivatives and the composition law of the action, and $U_{e_G} = \text{Id}$ by definition. $\square$

We now define the transported kernels and operators.

Definition 4.4 (Transported kernel and operator). *For $g \in G$, define a kernel $K_g : E \times E \rightarrow \mathbb{R}$ by*

$$K_g(x, y) = J_g(x)^{1/2} J_g(y)^{1/2} K_e(g^{-1} \cdot x, g^{-1} \cdot y), \quad (7)$$

and the associated integral operator $T_g : L^2(\mu) \rightarrow L^2(\mu)$ by

$$(T_g f)(x) = \int_E K_g(x, y) f(y) \mu(dy). \quad (8)$$

The expression (7) is the analytic incarnation of the twin transport of the base kernel K_e under the action of G .

Lemma 4.5 (Conjugation identity). *For each $g \in G$ and $f \in L^2(\mu)$,*

$$T_g = U_g T_e U_g^{-1}, \quad (9)$$

and in particular T_g is self-adjoint, positive and compact.

Proof. Fix $g \in G$ and $f \in L^2(\mu)$. Using (6) and (3), we compute

$$(U_g^{-1}f)(y) = J_{g^{-1}}(y)^{1/2} f(g \cdot y),$$

so that

$$(T_e U_g^{-1}f)(z) = \int_E K_e(z, y) J_{g^{-1}}(y)^{1/2} f(g \cdot y) \mu(dy).$$

Applying U_g and using (6) again gives

$$(U_g T_e U_g^{-1}f)(x) = J_g(x)^{1/2} \int_E K_e(g^{-1} \cdot x, y) J_{g^{-1}}(y)^{1/2} f(g \cdot y) \mu(dy).$$

Perform the change of variables $y = g^{-1} \cdot u$ (so $u = g \cdot y$). As in the proof of Lemma 4.3, $\int \Phi(y) \mu(dy) = \int \Phi(g^{-1} \cdot u) J_g(u) \mu(du)$, i.e. the measure factor is

$$\mu(dy) = J_g(u) \mu(du), \quad \text{while} \quad J_{g^{-1}}(g^{-1} \cdot u) = J_g(u)^{-1}$$

(the latter from the cocycle $J_g(u) J_{g^{-1}}(g^{-1} \cdot u) = J_e(u) = 1$). Substituting, the two factors combine as $J_{g^{-1}}(g^{-1} \cdot u)^{1/2} J_g(u) = J_g(u)^{-1/2} J_g(u) = J_g(u)^{1/2}$, so

$$\begin{aligned} (U_g T_e U_g^{-1}f)(x) &= J_g(x)^{1/2} \int_E K_e(g^{-1} \cdot x, g^{-1} \cdot u) J_{g^{-1}}(g^{-1} \cdot u)^{1/2} f(u) J_g(u) \mu(du) \\ &= J_g(x)^{1/2} \int_E K_e(g^{-1} \cdot x, g^{-1} \cdot u) J_g(u)^{1/2} f(u) \mu(du). \end{aligned}$$

Comparing with (8) and (7), we see that

$$(U_g T_e U_g^{-1}f)(x) = \int_E K_g(x, u) f(u) \mu(du) = (T_g f)(x),$$

which proves (9). Self-adjointness, positivity and compactness follow from the corresponding properties of T_e and the fact that U_g is unitary. $\square$

4.2 Spectral Equivariance Theorem

We are now ready to state and prove the main result of this section.

Theorem 4.6 (Spectral Equivariance Theorem). *Assume that T_e admits the spectral decomposition (5) with eigenpairs $(\lambda_k, P_k)_{k \geq 0}$, where $\{P_k\}_{k \geq 0}$ is an orthonormal basis of $L^2(\mu)$. For each $g \in G$, define*

$$P_k^{(g)} = U_g P_k, \quad k \geq 0.$$

Then:

- (a) *For every $g \in G$, $\{P_k^{(g)}\}_{k \geq 0}$ is an orthonormal basis of $L^2(\mu)$.*
- (b) *For every $g \in G$ and $k \geq 0$,*

$$T_g P_k^{(g)} = \lambda_k P_k^{(g)}.$$

In particular, the spectrum of T_g coincides with that of T_e .

(c) The transported kernel K_g admits the spectral expansion

$$K_g(x, y) = \sum_{k \geq 0} \lambda_k P_k^{(g)}(x) P_k^{(g)}(y) \quad \text{in } L^2(\mu \otimes \mu). \quad (10)$$

Proof. (a) Unitarity of U_g (Lemma 4.3) implies that for all $k, \ell \geq 0$,

$$\langle P_k^{(g)}, P_\ell^{(g)} \rangle = \langle U_g P_k, U_g P_\ell \rangle = \langle P_k, P_\ell \rangle = \delta_{k\ell}.$$

Moreover, since $\{P_k\}_{k \geq 0}$ is an orthonormal basis and U_g is surjective (unitary), the family $\{P_k^{(g)}\}_{k \geq 0}$ is again an orthonormal basis of $L^2(\mu)$.

(b) Using the conjugation identity (9) and the eigen-relation (4), we compute

$$T_g P_k^{(g)} = U_g T_e U_g^{-1} (U_g P_k) = U_g T_e P_k = U_g (\lambda_k P_k) = \lambda_k U_g P_k = \lambda_k P_k^{(g)}.$$

Thus $P_k^{(g)}$ is an eigenfunction of T_g with eigenvalue λ_k .

(c) Since $\{P_k^{(g)}\}_{k \geq 0}$ is a complete orthonormal basis of $L^2(\mu)$ and T_g is self-adjoint, we have the usual spectral representation

$$T_g f = \sum_{k \geq 0} \lambda_k \langle f, P_k^{(g)} \rangle P_k^{(g)}, \quad f \in L^2(\mu),$$

with convergence in $L^2(\mu)$. In kernel form, this is equivalent to

$$K_g(x, y) = \sum_{k \geq 0} \lambda_k P_k^{(g)}(x) P_k^{(g)}(y) \quad \text{in } L^2(\mu \otimes \mu),$$

which is precisely (10). $\square$

Remark 4.7 (Orthogonal polynomials in twin spaces). *In the classical situation where μ is supported on an interval of $\mathbb{R}$ and $\{P_k\}_{k \geq 0}$ are orthonormal polynomials with respect to μ , the functions $P_k^{(g)}$ can often be identified with orthogonal polynomial systems in the transformed coordinates $g \cdot x$, possibly up to multiplicative Jacobian factors. In this sense, Theorem 4.6 shows that orthogonal polynomial bases in different twin spaces are obtained by transporting a single base system $\{P_k\}$ through the group action.*

Corollary 4.8 (Equivariance of spectral truncation). *For any $K \in \mathbb{N}$ and $g \in G$, define the spectral truncation operators*

$$S_{e,K} f = \sum_{k=0}^K \langle f, P_k \rangle P_k, \quad S_{g,K} f = \sum_{k=0}^K \langle f, P_k^{(g)} \rangle P_k^{(g)}.$$

Then, for all $f \in L^2(\mu)$,

$$S_{g,K} = U_g S_{e,K} U_g^{-1}.$$

In particular, spectral truncation is equivariant under twin transport.

Proof. Using $P_k^{(g)} = U_g P_k$ and the unitarity of U_g , we have

$$\langle f, P_k^{(g)} \rangle = \langle f, U_g P_k \rangle = \langle U_g^{-1} f, P_k \rangle.$$

Therefore

$$S_{g,K} f = \sum_{k=0}^K \langle f, P_k^{(g)} \rangle P_k^{(g)} = \sum_{k=0}^K \langle U_g^{-1} f, P_k \rangle U_g P_k = U_g \left(\sum_{k=0}^K \langle U_g^{-1} f, P_k \rangle P_k \right) = U_g S_{e,K} U_g^{-1} f,$$

which yields the desired identity. $\square$

Corollary 4.9 (Spectral Equivariance for Classical Orthogonal Polynomials). *Let (E, μ) be either $([-1, 1], dx)$ or $(\mathbb{R}, \gamma)$ where γ is the standard Gaussian measure. Let*

$$K_e(x, y) = \sum_{k \geq 0} \lambda_k P_k(x) P_k(y)$$

be the classical Mercer expansion of the base kernel in the corresponding orthogonal polynomial basis:

- Legendre polynomials P_k on $[-1, 1]$ with weight $w(x) \equiv 1$,
- Hermite polynomials H_k on $\mathbb{R}$ with respect to $d\gamma(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx$.

Let G act on E by:

$$g \cdot x = \begin{cases} ax + b, & \text{Legendre case,} \\ \alpha x, & \text{Hermite case.} \end{cases}$$

Then for each $g \in G$, the transported kernel

$$K_g(x, y) = J_g(x)^{1/2} J_g(y)^{1/2} K_e(g^{-1} \cdot x, g^{-1} \cdot y)$$

admits the spectral expansion

$$K_g(x, y) = \sum_{k \geq 0} \lambda_k P_k^{(g)}(x) P_k^{(g)}(y),$$

where:

- in the Legendre case, the transported basis $P_k^{(g)}$ is a normalized affine transform of P_k ;
- in the Hermite case, the transported basis is

$$H_k^{(g)}(x) = \alpha^{-1/2} \exp\left(\frac{1}{4}(1 - \alpha^{-2})x^2\right) H_k(\alpha^{-1}x),$$

forming again an orthonormal basis of $L^2(\gamma)$.

Proof. The Legendre case corresponds to the affine group acting on $[-1, 1]$ with a constant Jacobian $J_{a,b} \equiv a^{-1}$. The Hermite case corresponds to the multiplicative group $(0, \infty)$ acting on $\mathbb{R}$ by dilations $x \mapsto \alpha x$; the pushforward of $\gamma = N(0, 1)$ is $N(0, \alpha^2)$, so the Radon–Nikodym derivative (Gaussian Jacobian) is

$$J_\alpha(x) = \frac{d(\tau_{\alpha*}\gamma)}{d\gamma}(x) = \alpha^{-1} \exp\left(\frac{1}{2}(1 - \alpha^{-2})x^2\right).$$

In both cases, the assumptions of Theorem 4.6 hold: U_g is unitary on $L^2(\mu)$, $T_g = U_g T_e U_g^{-1}$, and $P_k^{(g)} := U_g P_k$ forms an orthonormal basis that diagonalizes T_g . The explicit formula for $H_k^{(g)}$ then follows from

$$(U_\alpha f)(x) = J_\alpha(x)^{1/2} f(\alpha^{-1}x) = \alpha^{-1/2} \exp\left(\frac{1}{4}(1 - \alpha^{-2})x^2\right) f(\alpha^{-1}x),$$

applied to $f = H_k$. (Orthonormality is automatic from unitarity of U_α ; one checks directly that $\int |H_k^{(g)}|^2 d\gamma = \int |H_k|^2 d\gamma = 1$.) $\square$

5 Orthogonal Series Estimators as Projections in Twin Kernel Spaces

In this section we show that classical orthogonal polynomial estimators arise naturally as orthogonal projections in Twin Kernel Spaces. The analysis relies crucially on the Spectral Equivariance Theorem (Theorem 4.6) proved previously. The key idea is that an estimator constructed in a twin space $g \cdot E$ is obtained by transporting the base estimator in E through the unitary operator U_g . This gives a unified geometric interpretation of orthogonal series methods across all twin spaces.

5.1 Projection in the base RKHS

Let K_e be the base kernel with spectral expansion

$$K_e(x, y) = \sum_{k \geq 0} \lambda_k P_k(x) P_k(y).$$

Let $\mathcal{H}_e$ be the reproducing kernel Hilbert space (RKHS) associated with K_e . Given a target function $f \in L^2(\mu)$ (e.g., a density or regression function) and a truncation level $K \in \mathbb{N}$, the classical orthogonal series estimator is

$$\hat{f}_{e,K}(x) = \sum_{k=0}^K \hat{\theta}_k P_k(x), \quad \hat{\theta}_k = \langle \hat{f}_{\text{emp}}, P_k \rangle, \quad (11)$$

where $\hat{f}_{\text{emp}}$ is the empirical estimator (histogram, empirical measure, or empirical regression operator depending on the setting).

Proposition 5.1 (Projection property in the base twin space). *The estimator $\hat{f}_{e,K}$ is the orthogonal projection of $\hat{f}_{\text{emp}}$ onto the subspace*

$$\mathcal{H}_{e,K} = \text{span}\{P_0, \dots, P_K\} \subset L^2(\mu),$$

i.e.,

$$\hat{f}_{e,K} = S_{e,K} \hat{f}_{\text{emp}}.$$

Proof. Since $\{P_k\}$ is an orthonormal basis, $S_{e,K}$ is exactly the $L^2(\mu)$ -projection onto $\mathcal{H}_{e,K}$. Equation (11) is the standard expression for the orthogonal projection in an ONB, proving the claim. $\square$

Thus the classical construction is a projection estimator in the base twin space.

5.2 Transport of estimators under group actions

We now describe how estimation in a transported geometry $g \cdot E$ is obtained. Given the group action and the unitary operator U_g , define the transported empirical estimator

$$\hat{f}_{\text{emp}}^{(g)} = U_g \hat{f}_{\text{emp}}.$$

The transported orthonormal basis is

$$P_k^{(g)} = U_g P_k,$$

and the truncated space is

$$\mathcal{H}_{g,K} = \text{span}\{P_0^{(g)}, \dots, P_K^{(g)}\}.$$

Proposition 5.2 (Projection in transported twin spaces). *For each $g \in G$,*

$$\hat{f}_{g,K} = S_{g,K} \hat{f}_{\text{emp}}^{(g)}$$

is the $L^2(\mu)$ -orthogonal projection of the empirical estimator onto $\mathcal{H}_{g,K}$.

Proof. Since $\{P_k^{(g)}\}$ is an orthonormal basis (Theorem 4.6), the projection onto their span is exactly

$$S_{g,K} f = \sum_{k=0}^K \langle f, P_k^{(g)} \rangle P_k^{(g)}.$$

Substituting $f = \hat{f}_{\text{emp}}^{(g)}$ yields the expression above. $\square$

Thus estimation in any twin space corresponds to projection in a geometry determined by the transported kernel K_g .

5.3 Equivariance of orthogonal series estimators

We now relate $\hat{f}_{e,K}$ and $\hat{f}_{g,K}$.

Theorem 5.3 (Equivariance of spectral estimators). *For every $g \in G$ and truncation level K ,*

$$\hat{f}_{g,K} = U_g \hat{f}_{e,K}.$$

Proof. By Corollary 4.8, we have $S_{g,K} = U_g S_{e,K} U_g^{-1}$. Applying this to $\hat{f}_{\text{emp}}^{(g)} = U_g \hat{f}_{\text{emp}}$ gives

$$\hat{f}_{g,K} = S_{g,K} \hat{f}_{\text{emp}}^{(g)} = U_g S_{e,K} U_g^{-1} (U_g \hat{f}_{\text{emp}}) = U_g S_{e,K} \hat{f}_{\text{emp}} = U_g \hat{f}_{e,K}.$$

$\square$

This theorem reveals that orthogonal polynomial estimators satisfy a strong transformation rule: **they commute with the twin transport induced by the group action.**

5.4 Geometric interpretation

Theorem 5.3 implies the following:

- The same statistical procedure (spectral truncation) produces consistent estimators in all twin spaces.
- Choice of geometry (via g) modifies the basis functions but does not modify the form of the estimator.
- Estimation can be viewed as occurring in different “reference frames” of the kernel geometry.
- Orthogonal polynomials in each twin space arise from transporting the base orthogonal polynomials.

Thus the Twin Kernel viewpoint shows that the apparent diversity of orthogonal polynomial bases (Legendre, Chebyshev, Hermite, Laguerre, Jacobi, etc.) reflects different geometric frames rather than fundamentally different estimators.

5.5 Example: Hermite estimation in a rescaled Gaussian geometry

Let $g_\alpha : x \mapsto \alpha x$ with $\alpha > 0$. Using Corollary 4.9,

$$P_k^{(g_\alpha)}(x) = H_k^{(g_\alpha)}(x) = U_\alpha H_k(x).$$

The estimator in the transported twin space is

$$\hat{f}_{g_\alpha, K}(x) = \sum_{k=0}^K \langle U_\alpha \hat{f}_{\text{emp}}, H_k^{(g_\alpha)} \rangle H_k^{(g_\alpha)}(x).$$

By the Equivariance Theorem:

$$\hat{f}_{g_\alpha, K} = U_\alpha \hat{f}_{e, K}.$$

Thus Hermite estimation in a “stretched” Gaussian geometry corresponds exactly to transporting the base Hermite estimator. This provides a fully explicit operational interpretation of geometric deformation in Gaussian twin spaces.

5.6 Implication for nonparametric estimation

The analysis shows:

- Orthogonal polynomial estimators are geometrically coherent across all kernel-induced twin spaces.
- The choice of truncation K is stable under twin transport.
- Statistical properties (risk, bias-variance tradeoff, minimax rates) are preserved under transport.

Hence Twin Kernel Spaces provide a natural geometric unification of orthogonal series methods and enable their extension to settings where the underlying geometry undergoes transformations.

6 Relationship with Classical Kernel Estimators

This section explains how traditional kernel estimators fit naturally into the Twin Kernel Space framework. We show that classical smoothing kernels correspond to transported versions of a base kernel, and that the widely used estimators of Parzen–Rosenblatt and Nadaraya–Watson can be interpreted as projections in a twin geometry. This clarifies the relationship between orthogonal series estimation and kernel smoothing and reveals that both methods arise from the same geometric mechanism: spectral truncation or spectral attenuation in a transported RKHS.

6.1 Kernel estimators as RKHS projections

Let K_h be a classical smoothing kernel with bandwidth $h > 0$, such as the Gaussian kernel

$$K_h(x, y) = \frac{1}{h} K\left(\frac{x - y}{h}\right),$$

or any translation-invariant kernel of the form $K_h(x, y) = K((x - y)/h)/h$.

Define the estimator

$$\hat{f}_h(x) = \int_E K_h(x, y) d\hat{F}_n(y), \tag{12}$$

where $\hat{F}_n$ is the empirical distribution.

It is well known (e.g., via the representer theorem or the reproducing property [Wahba, 1990, Berlinet and Thomas-Agnan, 2004]) that $\hat{f}_h$ equals the projection of $\hat{f}_{\text{emp}}$ onto the RKHS associated with K_h :

$$\hat{f}_h = \Pi_{K_h}(\hat{f}_{\text{emp}}),$$

where Π_{K_h} is the orthogonal projection onto the RKHS $\mathcal{H}_{K_h}$.

Thus kernel estimators are RKHS projections, just as orthogonal polynomial estimators are. The question is: *how does K_h relate to the twin kernel K_g ?*

The answer is that **bandwidth scaling corresponds to group action and kernel transport**.

6.2 Bandwidth as a group action: $x \mapsto x/h$

Consider the multiplicative group $G = (0, \infty)$ acting on $E = \mathbb{R}$ (with Lebesgue base measure) via

$$g_h \cdot x = hx, \quad \text{so that} \quad g_h^{-1} \cdot x = \frac{x}{h}.$$

The Jacobian is $J_h(x) = h^{-1}$, and, since $K_{g_h}(x, y) = J_h(x)^{1/2} J_h(y)^{1/2} K_e(g_h^{-1}x, g_h^{-1}y)$, the transported kernel is

$$K_{g_h}(x, y) = h^{-1} K_e\left(\frac{x}{h}, \frac{y}{h}\right).$$

Thus, a classical smoothing kernel K_h is nothing but a transported version of the base kernel:

$$K_h = K_{g_h}.$$

In other words:

Bandwidth selection = choice of twin space geometry.

Proposition 6.1. *The Parzen–Rosenblatt estimator $\hat{f}_h$ is the orthogonal projection of $\hat{f}_{\text{emp}}$ onto the twin RKHS $\mathcal{H}_{g_h}$ associated with the transported kernel K_{g_h} .*

Proof. By construction, $K_h = K_{g_h}$, and (12) is exactly the RKHS projection of $\hat{f}_{\text{emp}}$ onto $\mathcal{H}_{g_h}$. $\square$

Thus kernel smoothing is a projection operation in a transported geometry.

6.3 Spectral interpretation: attenuation vs truncation

Let K_e admit the Mercer expansion

$$K_e(x, y) = \sum_{k \geq 0} \lambda_k P_k(x) P_k(y),$$

and let K_{g_h} be the transported kernel:

$$K_{g_h}(x, y) = \sum_{k \geq 0} \lambda_k(h) P_k^{(g_h)}(x) P_k^{(g_h)}(y).$$

Classical kernel smoothing corresponds to an estimator of the form

$$\hat{f}_h = \sum_{k \geq 0} \lambda_k(h) \langle \hat{f}_{\text{emp}}, P_k^{(g_h)} \rangle P_k^{(g_h)}.$$

Spectral smoothing. For translation-invariant kernels $K_h(x, y) = K((x - y)/h)/h$, one has

$$\lambda_k(h) = \lambda_k \exp(-c_k h^2),$$

which produces *spectral attenuation*.

Orthogonal series. For polynomial estimators,

$$\hat{f}_{e,K} = \sum_{k=0}^K \langle \hat{f}_{\text{emp}}, P_k \rangle P_k,$$

which corresponds to *hard spectral truncation*.

Thus:

Kernel estimation = soft spectral cutoff, Orthogonal series = hard spectral cutoff.
--

Both are spectral operations in twin RKHSs.

6.4 Equivariance of kernel estimators

We now show that kernel estimators satisfy the same equivariance principle as orthogonal polynomial estimators.

Theorem 6.2 (Equivariance of kernel smoothing). *Let $K_h = K_{g_h}$ be a transported kernel. Then:*

$$\hat{f}_h = U_{g_h} \hat{f}_e,$$

where $\hat{f}_e$ is the projection associated with the base kernel K_e .

Proof. Since $T_h = U_{g_h} T_e U_{g_h}^{-1}$, the projection operators satisfy

$$\Pi_{K_h} = U_{g_h} \Pi_{K_e} U_{g_h}^{-1}.$$

Applying this to $\hat{f}_{\text{emp}}$ yields

$$\hat{f}_h = \Pi_{K_h} \hat{f}_{\text{emp}} = U_{g_h} \Pi_{K_e} U_{g_h}^{-1} \hat{f}_{\text{emp}} = U_{g_h} \Pi_{K_e} \hat{f}_{\text{emp}} = U_{g_h} \hat{f}_e.$$

□

Thus *kernel estimation is itself an equivariant statistical operation* in Twin Kernel Spaces.

6.5 Comparison table: orthogonal series vs. kernel smoothing

Method	Spectral effect	Twin geometry interpretation
Orthogonal series	hard cutoff $\lambda_k^{\text{new}} = \lambda_k \mathbf{1}_{k \leq K}$	projection in $\mathcal{H}_e$
Kernel smoothing	soft cutoff $\lambda_k^{\text{new}} = \lambda_k \phi_k(h)$	projection in $\mathcal{H}_{g_h}$
Local polynomial	hybrid attenuation + local geometry	spatially varying twin space
Spline smoothing	penalized spectral cutoff	differential twin space

Table 1: Comparison of nonparametric methods in the Twin Kernel Space framework.

This demonstrates that a wide variety of nonparametric estimators are instances of the same principle: projection in a transported RKHS.

6.6 Conclusion: kernel smoothing as Twin Kernel transport

We can summarize the relationship as follows:

- Selecting a bandwidth h corresponds to choosing a twin space $g_h \cdot E$.
- Kernel estimators are projections in the twin RKHS $\mathcal{H}_{g_h}$.
- Orthogonal and kernel estimators differ only in their spectral filters.
- All these estimators satisfy geometric equivariance.

This places classical kernel estimation and orthogonal polynomial estimation in a single geometric framework and establishes Twin Kernel Spaces as a natural language for spectral non-parametric inference.

7 Examples and Illustrations

We illustrate the framework by examining the two most classical orthogonal polynomial systems (Legendre and Hermite), and then by presenting a multimodal example showing how kernel transport reveals geometric deformations in the underlying data-generating process.

7.1 Gaussian Twin Spaces and Hermite Polynomials

Let $E = \mathbb{R}$ with Gaussian measure $d\gamma(x) = \frac{1}{\sqrt{2\pi}} e^{-x^2/2} dx$. The base kernel is the Mehler kernel [Mehler, 1866]

$$K_e(x, y) = \sum_{k=0}^{\infty} \rho^k H_k(x) H_k(y), \quad |\rho| < 1,$$

whose eigenfunctions are the Hermite polynomials.

Consider the dilation action

$$g_\alpha : x \mapsto \alpha x, \quad \alpha > 0.$$

The Jacobian (Radon–Nikodym derivative of the pushforward $N(0, \alpha^2)$ with respect to $\gamma = N(0, 1)$) is

$$J_\alpha(x) = \alpha^{-1} \exp\left(\frac{1}{2}(1 - \alpha^{-2})x^2\right),$$

and the transported kernel is

$$K_{g_\alpha}(x, y) = J_\alpha(x)^{1/2} J_\alpha(y)^{1/2} K_e(\alpha^{-1}x, \alpha^{-1}y).$$

By Corollary 4.9, the transported basis is

$$H_k^{(g_\alpha)}(x) = U_\alpha H_k(x) = \alpha^{-1/2} \exp\left(\frac{1}{4}(1 - \alpha^{-2})x^2\right) H_k(\alpha^{-1}x).$$

Thus the spectral expansion in the twin space is

$$K_{g_\alpha}(x, y) = \sum_{k=0}^{\infty} \rho^k H_k^{(g_\alpha)}(x) H_k^{(g_\alpha)}(y).$$

Interpretation. The geometry becomes “stretched”: small α emphasizes fine-scale oscillations, large α smooths them. The deformed Hermite basis reflects this modified geometry.

7.2 Affine Twin Spaces and Legendre Polynomials

Let $E = [-1, 1]$ with Lebesgue measure, and P_k the orthonormal Legendre polynomials. Consider the affine transformation

$$g_{a,b} : x \mapsto ax + b, \quad a > 0, \quad b \in \mathbb{R},$$

with $a[-1, 1] + b \subset [-1, 1]$.

The Jacobian is $J_{a,b}(x) = a^{-1}$, giving the transported kernel

$$K_{g_{a,b}}(x, y) = a^{-1} K_e\left(\frac{x-b}{a}, \frac{y-b}{a}\right).$$

By the spectral equivariance theorem,

$$P_k^{(g_{a,b})}(x) = a^{-1/2} P_k\left(\frac{x-b}{a}\right).$$

The Legendre expansion becomes

$$K_{g_{a,b}}(x, y) = \sum_{k \geq 0} \lambda_k P_k^{(g_{a,b})}(x) P_k^{(g_{a,b})}(y),$$

with identical eigenvalues.

Interpretation. Affine geometric deformations correspond exactly to transporting the Legendre basis. The polynomial oscillation pattern is preserved, but shifted and rescaled.

7.3 Multimodal Transport Example: Adapting Spectral Bases to Complex Geometry

We now illustrate a phenomenon for which Twin Kernel Spaces are particularly well suited: non-stationary or multimodal data.

Consider a density consisting of two Gaussian modes:

$$f(x) = 0.5 \varphi(x+2) + 0.5 \varphi(x-2), \quad \varphi \text{ standard normal.}$$

In the base space $E = \mathbb{R}$, the Hermite basis is poorly adapted to this shape: it oscillates at the wrong spatial locations and requires many terms for an accurate approximation.

However, let $g_1 : x \mapsto x+2$ and $g_2 : x \mapsto x-2$. Define the transported bases

$$H_k^{(g_1)}(x) = U_{g_1} H_k(x), \quad H_k^{(g_2)}(x) = U_{g_2} H_k(x),$$

which correspond to translating the entire geometry around each mode.

Key observation. The multimodal density admits a much more efficient representation by combining expansions in several twin spaces:

$$f(x) \approx 0.5 \sum_{k=0}^K \theta_{k,1} H_k^{(g_1)}(x) + 0.5 \sum_{k=0}^K \theta_{k,2} H_k^{(g_2)}(x),$$

with dramatically fewer coefficients.

This is because each transported twin space “centers” the geometry at a mode of the density.

Interpretation. The spectral equivariance property means that orthogonal polynomials transported by group actions adapt naturally to complex geometries. Twin Kernel Spaces thus provide a principled geometric mechanism for:

- mode-adaptive estimation,
- spatially varying spectral representations,
- mixture-aware kernels.

This example shows that twin transport can be combined with spectral estimation to obtain geometry-adaptive nonparametric estimators.

8 Convergence in Twin Kernel Spaces

We study the convergence of spectral estimators in Twin Kernel Spaces. The key feature is that convergence in any transported geometry $g \cdot E$ is equivalent to convergence in the base geometry E , due to unitarity of the transport operator U_g .

8.1 Assumptions

Let K_e be a base Mercer kernel with eigenvalues $(\lambda_k)_{k \geq 0}$ and orthonormal eigenfunctions $(P_k)_{k \geq 0}$. Assume:

- (A1) **Spectral decay:** $\lambda_k \asymp k^{-2s}$ or $\lambda_k \asymp e^{-ck^\alpha}$ for some $s > 0$ or $\alpha > 0$.
- (A2) **Smoothness of f :** $f \in \mathcal{H}_e^t := \{f = \sum \theta_k P_k : \sum \theta_k^2 \lambda_k^{-t} < \infty\}$ for some $t > 0$.
- (A3) **i.i.d. sampling:** The empirical estimator $\hat{f}_{\text{emp}}$ satisfies $\|\hat{f}_{\text{emp}} - f\|_{L^2(\mu)} = O_p(n^{-1/2})$.

The Twin Kernel Space associated with $g \in G$ has eigenfunctions $(P_k^{(g)}) = (U_g P_k)$ and identical eigenvalues.

8.2 Main convergence result

Theorem 8.1 (Convergence of spectral estimators in Twin Kernel Spaces). *Let K be a truncation level such that $K = K(n) \rightarrow \infty$ and $K(n)/n \rightarrow 0$. For every $g \in G$, the estimator*

$$\hat{f}_{g,K} = S_{g,K} \hat{f}_{\text{emp}}^{(g)}$$

satisfies

$$\|\hat{f}_{g,K} - f^{(g)}\|_{L^2(\mu)} \xrightarrow{p} 0,$$

and the convergence rate is identical to that in the base geometry:

$$\|\hat{f}_{g,K} - f^{(g)}\|_{L^2(\mu)} \asymp \|\hat{f}_{e,K} - f\|_{L^2(\mu)}.$$

Proof. Using $P_k^{(g)} = U_g P_k$ and unitarity,

$$\|\hat{f}_{g,K} - f^{(g)}\| = \|U_g(\hat{f}_{e,K} - f)\| = \|\hat{f}_{e,K} - f\|.$$

Consistency and the rate follow from the classical bias–variance decomposition [Tsybakov, 2009]. $\square$

Thus, **convergence is geometry-invariant**: transporting the kernel does not change asymptotic behavior.

9 Bias–Variance Analysis in Twin Kernel Spaces

We now compute the bias and variance components of the estimator

$$\hat{f}_{g,K} = \sum_{k=0}^K \hat{\theta}_k^{(g)} P_k^{(g)}.$$

Recall that

$$\hat{\theta}_k^{(g)} - \theta_k^{(g)} = \langle \hat{f}_{\text{emp}}^{(g)} - f^{(g)}, P_k^{(g)} \rangle.$$

9.1 Bias

Since U_g preserves smoothness,

$$f^{(g)} = U_g f = \sum_{k \geq 0} \theta_k P_k^{(g)} \quad \text{with } \theta_k = \langle f, P_k \rangle.$$

The bias term is

$$\|\mathbb{E} \hat{f}_{g,K} - f^{(g)}\|^2 = \sum_{k > K} \theta_k^2.$$

If f lies in a Sobolev-type ball $\mathcal{H}_e^t$,

$$\sum_{k > K} \theta_k^2 = O(K^{-2t}),$$

and therefore

$$\text{Bias}_K \asymp K^{-t}.$$

9.2 Variance

The variance term is

$$\mathbb{E} \|\hat{f}_{g,K} - \mathbb{E} \hat{f}_{g,K}\|^2 = \sum_{k=0}^K \text{Var}(\hat{\theta}_k^{(g)}).$$

Under standard regularity,

$$\text{Var}(\hat{\theta}_k^{(g)}) = \frac{1}{n} \sigma_k^2, \quad \sum_{k=0}^K \sigma_k^2 \lesssim K.$$

Hence

$$\text{Var}_K \asymp \frac{K}{n}.$$

9.3 Optimal choice of K

Balancing bias and variance:

$$K^{-2t} \asymp \frac{K}{n} \quad \Rightarrow \quad K \asymp n^{1/(2t+1)}.$$

Thus the optimal rate is

$$\|\hat{f}_{g,K} - f^{(g)}\| \asymp n^{-t/(2t+1)},$$

the classical minimax rate [Stone, 1982, Tsybakov, 2009].

Key message.

Bias–variance tradeoff is invariant under twin transport.

10 Minimax Rates in Twin Kernel Spaces

We now extend the previous results to minimax lower bounds.

10.1 Function classes

Let

$$\mathcal{F}_t(R) = \left\{ f = \sum_{k \geq 0} \theta_k P_k \mid \sum_{k \geq 0} \theta_k^2 \lambda_k^{-t} \leq R^2 \right\}.$$

Transport defines identical classes:

$$\mathcal{F}_t^{(g)}(R) = \{f^{(g)} = U_g f : f \in \mathcal{F}_t(R)\}.$$

Theorem 10.1 (Minimax risk equivalence in twin spaces). *Let $\hat{f}$ be any estimator. Then for every $g \in G$,*

$$\inf_{\hat{f}} \sup_{f \in \mathcal{F}_t(R)} \mathbb{E} \|\hat{f}^{(g)} - f^{(g)}\|^2 = \inf_{\hat{f}} \sup_{f \in \mathcal{F}_t(R)} \mathbb{E} \|\hat{f} - f\|^2.$$

In particular,

$$\mathcal{R}_n^* \asymp n^{-2t/(2t+1)}$$

in all twin spaces.

Proof. Since U_g is unitary,

$$\|U_g(\hat{f} - f)\| = \|\hat{f} - f\|,$$

and

$$U_g(\mathcal{F}_t(R)) = \mathcal{F}_t^{(g)}(R).$$

Apply standard least favorable priors and Le Cam's method [Le Cam, 1986, Yu, 1997]; the geometry does not affect the lower bound. $\square$

Thus **minimax risk is entirely geometry-invariant**.

Remark 10.2 (Why the spectral estimator is exactly minimax, with no logarithmic factor). *The truncated series estimator $\hat{f}_{g,K}$ is a projection estimator: it fits the K leading coefficients directly, so its variance is the metric complexity K/n of the model and the bias-variance balance of Section 10 gives the exact rate $n^{-2t/(2t+1)}$, attaining the lower bound of Theorem 10.1 up to constants—no logarithmic factor. This is the structural reason a spectral/orthogonal-series method is sharper than a description-length (MDL) procedure on the same sieve: the latter must quantize each coefficient into a prefix code, paying an extra $\log(1/\delta) \asymp \log n$ per coefficient and so attaining only the near-minimax rate $(\log n/n)^{2t/(2t+1)}$. Choosing K by a penalty $\text{pen}(K) \asymp K/n$ (rather than a code length $\asymp K \log n$) preserves the exact rate adaptively in the smoothness t ; the geometric invariance above carries this exact-rate optimality verbatim across all twin spaces.*

11 Conclusion and Future Work

11.1 Summary

We have developed a unified geometric framework for nonparametric estimation based on the notion of Twin Kernel Spaces. Our main contributions are:

1. A precise formulation of Twin Kernel Spaces as orbits of reproducing kernels under group actions, providing a minimal yet powerful geometric structure.
2. The Spectral Equivariance Theorem (Theorem 4.6), establishing that eigenfunctions of transported kernels are obtained by unitary transport of base eigenfunctions, with spectra preserved.

3. A unified interpretation of orthogonal polynomial estimation and kernel smoothing as spectral operations (hard vs. soft truncation) in potentially different twin geometries.
4. Geometric invariance results showing that convergence rates, bias–variance tradeoffs, and minimax risks are preserved under twin transport.
5. Explicit worked examples demonstrating the framework for Hermite and Legendre polynomials, including applications to multimodal estimation.

11.2 Open problems

Several directions for future research emerge from this work:

1. **Adaptive twin space selection.** While we have shown that estimation is equivariant across twin spaces, the question of how to optimally select the twin space (or combination of twin spaces) for a given dataset remains open. Data-driven methods for twin space selection could lead to improved practical estimators.
2. **Multiscale twin spaces.** The multimodal example suggests that combinations of twin spaces at different scales could provide efficient representations. A systematic theory of multiscale twin decompositions would be valuable.
3. **Non-commutative group actions.** Our framework extends naturally to non-commutative groups, but the statistical implications have not been fully explored. Applications to rotation groups and more general Lie groups deserve investigation.
4. **Infinite-dimensional extensions.** Extension to function spaces beyond $L^2(\mu)$, including Banach spaces and spaces of distributions, would broaden applicability.
5. **Computational aspects.** Efficient algorithms for computing transported kernels and eigenfunctions in high-dimensional settings are needed for practical applications.

11.3 Connections to twin Hilbert spaces and physics

The terminology “twin” evokes connections to several areas of mathematics and physics:

- In operator theory, twin spaces appear in the study of invariant subspaces and model theory [Halmos, 1982, Nikolski, 2002].
- In quantum mechanics, related structures arise in the study of coherent states and symmetry groups [Perelomov, 1986].
- The equivariance properties we establish are reminiscent of gauge invariance in physics, where physical quantities are preserved under certain transformations.
- Connections to optimal transport theory [Villani, 2009] may provide additional geometric insights.

These connections suggest that Twin Kernel Spaces may have broader significance beyond statistics, and merit further investigation.

A Technical Appendices

A.1 Proof of auxiliary lemmas

Lemma A.1 (Concentration of spectral coefficients). *Let $\hat{f}_{\text{emp}}$ be the empirical estimator of a density or regression function. For each k ,*

$$|\langle \hat{f}_{\text{emp}} - f, P_k \rangle| = O_p(n^{-1/2}).$$

Uniformly for $k \leq K(n)$ with $K(n)/n \rightarrow 0$,

$$\sum_{k=0}^{K(n)} (\langle \hat{f}_{\text{emp}} - f, P_k \rangle)^2 = O_p(K/n).$$

Proof. Standard Bernstein inequalities for empirical processes [van der Vaart and Wellner, 1996]. $\square$

A.2 Spectral regularity

Lemma A.2 (Eigenvalue decay and Sobolev norm equivalence). *If $\lambda_k \asymp k^{-2s}$ then*

$$\|f\|_{\mathcal{H}_e^t}^2 \asymp \sum_{k \geq 0} k^{2st} \theta_k^2.$$

Proof. Direct from $\lambda_k^{-t} \asymp k^{2st}$. $\square$

A.3 Proof of minimax lower bound

Lemma A.3 (Fano lower bound in spectral spaces). *Let $\mathcal{F}_t(R)$ be as above. Then*

$$\inf_{\hat{f}} \sup_{f \in \mathcal{F}_t(R)} \mathbb{E} \|\hat{f} - f\|^2 \gtrsim n^{-2t/(2t+1)}.$$

Proof. Construct least favorable priors supported on linear combinations of P_k and use standard Fano inequalities [Yu, 1997, Tsybakov, 2009]. $\square$

A.4 Proof of Theorem 10.1

Proof. We combine Lemma A.3 with the identity

$$\|U_g(\hat{f} - f)\| = \|\hat{f} - f\|.$$

Transport preserves the entire spectral and geometric structure. $\square$

B Additional Kernel Identities

Lemma B.1 (Transport of translation-invariant kernels). *Let $K(x, y) = \psi(x - y)$. Under dilation $x \mapsto x/h$,*

$$K_{gh}(x, y) = h^{-1} \psi\left(\frac{x - y}{h}\right).$$

Proof. Direct substitution in the twin transport formula. $\square$

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